ET AI HACKATHON 2026

Problem Statements

Industrial | Smart Cities | Sustainability | Supply Chain | Geospatial | Public Safety | Sales Excellence

Each team must select one problem statement. Choose the challenge that best aligns with your team's skills, expertise, and interests.

1 AI-Powered Industrial Safety Intelligence for Zero-Harm Operations

Theme: *Industrial Intelligence / Worker Safety / Geospatial Safety Analytics*

PROBLEM CONTEXT

India's heavy industrial sector continues to pay a devastating human cost. According to DGFASLI, over 6,500 fatal workplace accidents were recorded in FY2023 — and that figure excludes most of the mining and construction sectors. In one of the most disturbing recent incidents, eight workers died at Visakhapatnam Steel Plant in January 2025 when entrapped gases triggered a sudden explosion in the coke oven battery — a facility that had functioning safety systems including gas detectors, permit-to-work controls, and SCADA. An investigation by The Wire found that warning signals from gas pressure sensors existed, but no intelligence layer connected those readings to operational decisions in time. This pattern — data present, but unacted upon — repeats itself across Indian heavy industry. A FICCI survey in 2024 found that over 60% of large industrial facilities rely on manual handoffs to coordinate between their own digital safety tools. The problem is not the absence of technology. It is the absence of a unified intelligence layer that fuses data from disparate sensors, shift logs, maintenance records, and video feeds into a real-time risk picture — and acts on it before a fatality, not after.

CHALLENGE STATEMENT

Build an AI-powered Industrial Safety Intelligence platform that brings together data from IoT sensors, SCADA systems, permit-to-work logs, CCTV feeds, and shift records into a single predictive layer. The system should detect compound risk conditions — like the co-occurrence of maintenance activity and hazardous gas accumulation — that no single sensor would flag alone, and trigger preemptive interventions before they escalate.

WHAT YOU MAY BUILD

Participants may explore areas such as:

- **Compound Risk Detection Engine —** Multi-agent system that correlates gas sensor readings, work permit activity, equipment maintenance status, and shift changeover patterns to identify dangerous combinations — such as confined space entry during abnormal process conditions — hours before they become critical.
- **Geospatial Safety Heatmap —** Real-time geospatial layer over plant layout that visualises risk zones dynamically as conditions change — integrating worker location data, hazardous area classifications, and active permit overlaps to give safety officers situational awareness across the entire facility.

- **Incident Pattern Intelligence —** RAG-powered agent that cross-references near-miss reports, historical incident data, and OISD/Factory Act regulatory guidance to identify recurring patterns that manual investigations miss — and surfaces them as actionable prevention priorities.
- **Digital Permit Intelligence Agent —** AI that analyses active permits against real-time plant conditions and flags dangerous simultaneous operations — for example, hot work permits issued in proximity to areas with elevated gas readings, a combination that has repeatedly preceded fatal incidents.
- **Emergency Response Orchestrator —** Autonomous agent that, on confirmed trigger, immediately initiates evacuation protocols, alerts response teams across channels, preserves sensor evidence, and generates a preliminary regulatory-compliant incident report — reducing the critical first 10 minutes from chaos to coordinated response.
- **Quality & Compliance Audit Agent —** AI layer that continuously monitors safety procedures, inspection records, and statutory compliance documentation against regulatory standards (OISD, DGMS, Factory Act) — flagging deviations before audits and generating corrective action workflows automatically.

These examples are illustrative only.

SUGGESTED TECHNOLOGIES

- Agentic AI / Multi-Agent Systems
- Geospatial Intelligence & Plant Layout Analytics
- RAG over incident and regulatory document corpora
- Computer Vision & CCTV Analytics
- IoT / SCADA Data Integration
- Knowledge Graphs (equipment-permit-risk relationships)

EXPECTED DELIVERABLES

- Working Prototype
- Architecture Diagram
- Presentation Deck
- Demo Video

Evaluation Focus Compound risk detection accuracy versus single-sensor baselines, prediction lead time before incident threshold, geospatial evidence quality, regulatory compliance coverage (OISD/Factory Act/DGMS), and demonstrated reduction in false negative rate — the metric that actually saves lives.

JUDGING CRITERIA

Criteria	Weight
Innovation	25%
Business Impact	25%
Technical Excellence	20%
Scalability	15%
User Experience	15%

2 AI-Driven Energy Supply Chain Resilience for Import-Dependent Economies

Theme: *Supply Chain Intelligence / Energy Security / Geopolitical Risk*

PROBLEM CONTEXT

India sources approximately 88% of its crude oil from imports, with 40–45% of that volume transiting through the Strait of Hormuz. This is not a theoretical risk — it is a structural vulnerability that geopolitical events repeatedly stress-test. The 2025 US-Iran standoff sent Brent crude up over 8% in a single session and forced Indian refiners onto spot markets at steep premiums. The situation has not stabilised: through early 2026, renewed US sanctions pressure on Iranian oil exports, maritime security incidents in the Persian Gulf, and escalating Houthi attacks on Red Sea shipping lanes have kept the supply threat live. India's Strategic Petroleum Reserves provide roughly 9.5 days of national consumption cover — a buffer that would be exhausted quickly in any sustained disruption. A McKinsey analysis of past energy supply shocks found that economies without automated rerouting and demand-management capability took an average of 47 days longer to stabilise supply than those with integrated response intelligence. Traditional supply chain planning tools were designed for predictable environments. They have no ability to model geopolitical scenario impacts in real time, dynamically evaluate alternative procurement corridors, or orchestrate coordinated response across refiners, logistics providers, and strategic reserves. That intelligence layer needs to be built.

CHALLENGE STATEMENT

Design an AI-powered Energy Supply Chain Resilience system that monitors geopolitical and logistics risk signals continuously, models disruption scenarios and their downstream economic impacts, and generates executable procurement rerouting recommendations — turning a reactive crisis response into a managed, anticipatory process.

WHAT YOU MAY BUILD

Participants may explore areas such as:

- **Geopolitical Risk Intelligence Agent —** Multi-source agent that ingests news feeds, shipping AIS data, sanctions registries, and commodity price signals to produce a live supply disruption probability score by corridor and supplier — updated continuously, not weekly.
- **Disruption Scenario Modeller —** AI system that simulates specific events — Hormuz partial closure, OPEC+ emergency cut, Red Sea shipping suspension — and computes cascading impacts on refinery run rates, domestic fuel prices, power sector stress, and GDP trajectory under each scenario.
- **Adaptive Procurement Orchestrator —** Agentic system that identifies and ranks alternative crude sources and logistics routes, factoring in spot market pricing, tanker availability, port congestion, and refinery grade compatibility — generating procurement recommendations that procurement teams can act on within hours.
- **Strategic Reserve Optimisation Agent —** AI layer that models optimal SPR drawdown schedules against supply gap forecasts, refinery demand curves, and replenishment window estimates — providing decision support to policymakers under time pressure.
- **Supply Chain Digital Twin —** Geospatial simulation of India's full energy supply network — from wellhead to refinery to distribution — that enables continuous 'what-if' analysis and serves as a persistent intelligence platform for supply chain resilience planning.

These examples are illustrative only.

SUGGESTED TECHNOLOGIES

- Agentic AI / Multi-Agent Systems
- Geospatial Intelligence (AIS vessel tracking, pipeline and port mapping)
- Predictive Analytics & Scenario Simulation
- RAG over geopolitical and commodity intelligence sources
- Knowledge Graphs (supplier-route-risk-refinery relationships)
- LLMs for policy and news signal extraction

EXPECTED DELIVERABLES

- Working Prototype
- Architecture Diagram
- Presentation Deck
- Demo Video

Evaluation Focus Disruption signal detection lead time and accuracy, quality and executability of procurement alternatives generated, scenario model fidelity (assumptions must be explicit and testable), geospatial evidence depth, and demonstrated end-to-end response time from signal to recommendation.

JUDGING CRITERIA

Criteria	Weight
Innovation	25%
Business Impact	25%
Technical Excellence	20%
Scalability	15%
User Experience	15%

3 AI for Industrial EV Supply Chain & Asset Intelligence: Accelerating Net Zero

Theme: *Sustainability / EV Manufacturing / Asset Performance Management / Supply Chain*

PROBLEM CONTEXT

India registered over 2 million electric vehicles in FY2025 — yet EVs still account for less than 7% of total vehicle sales, and in industrial and commercial segments (freight, mining, intra-plant logistics, construction equipment) penetration remains below 2.5%, per SIAM FY2025 data. The barriers are not primarily financial anymore — FAME-II has disbursed over Rs 10,000 crore in incentives, and TCO parity with diesel is approaching for many use cases. The real bottleneck is operational: industrial fleet operators lack the asset intelligence tools to manage EV procurement, battery lifecycle, charging infrastructure, and maintenance operations with the same rigour they apply to conventional equipment. Meanwhile, EV manufacturers themselves face significant supply chain complexity — battery-grade lithium, cobalt, and nickel sourcing, cell-to-pack quality traceability, and multi-tier supplier risk — that generic supply chain tools handle poorly. For India to meet its 30% EV penetration target for commercial vehicles by 2030, it needs an AI layer that addresses both the industrial adoption gap and the manufacturing supply chain gap simultaneously. This is fundamentally a problem of asset intelligence and supply chain visibility — the same disciplines that determine whether a steel plant or a refinery runs at optimal efficiency, applied to the emerging EV ecosystem.

CHALLENGE STATEMENT

Build an AI platform that addresses the industrial EV transition from two interconnected angles: (1) helping asset-intensive organisations manage EV fleets with the same operational rigour as conventional industrial equipment — covering procurement intelligence, predictive asset performance management, and maintenance operations; and (2) helping EV manufacturers manage the complex, quality-critical supply chains that make reliable EVs possible.

WHAT YOU MAY BUILD

Participants may explore areas such as:

- **EV Asset Performance Management (APM) Agent —** AI layer that monitors battery state-of-health, charging cycle patterns, thermal events, and degradation trajectories across a fleet — generating predictive maintenance triggers, optimal charge-discharge recommendations, and remaining useful life estimates per asset, analogous to APM for industrial machinery.
- **Fleet Electrification Readiness & Procurement Intelligence —** Agent that analyses existing fleet operational data (routes, payloads, duty cycles, dwell times) and maps each vehicle to its optimal EV replacement, with a confidence-scored transition readiness index and a procurement recommendation aligned to available OEM options and delivery lead times.
- **EV Supply Chain Risk & Traceability Agent —** Multi-agent system that tracks critical battery material supply chains (lithium, cobalt, NMC/LFP cells) across tiers — flagging supplier concentration risk, geopolitical exposure, quality deviations from incoming inspection data, and sustainability compliance gaps in a unified intelligence view.
- **Manufacturing Quality Intelligence (QMS Integration) —** AI quality agent for EV component manufacturing that correlates process parameters, incoming material quality data, and in-line inspection results to detect quality drift before defective product reaches assembly — supporting traceability from cell to pack to vehicle and enabling root-cause analysis across the supply chain.
- **Net Zero Progress & Carbon Intelligence Tracker —** Geospatial AI layer that tracks fleet electrification progress against organisational net zero commitments, quantifies Scope 1 and

3 emission reductions per route and asset class, and identifies the highest-impact next electrification priorities based on both carbon and operational criteria.

- **Maintenance Operations Optimiser —** AI agent that integrates EV fleet maintenance schedules, charging infrastructure uptime data, and workshop capacity to generate optimised maintenance plans — reducing unplanned downtime and ensuring charging infrastructure availability aligns with operational shift patterns.

These examples are illustrative only.

SUGGESTED TECHNOLOGIES

- Agentic AI / Asset Performance Management (APM) Integration
- Supply Chain Intelligence & Multi-Tier Risk Monitoring
- Geospatial Analytics (fleet routing, charging infrastructure planning)
- Quality Management System (QMS) Integration & SPC
- Predictive Analytics (battery degradation, maintenance forecasting)
- IoT / Telematics / BMS Data Integration
- Digital Twin for EV Fleet Operations

EXPECTED DELIVERABLES

- Working Prototype
- Architecture Diagram
- Presentation Deck
- Demo Video

Evaluation Focus Battery degradation prediction accuracy versus observed data, supply chain risk detection lead time, quality defect detection precision/recall in manufacturing use case, fleet electrification readiness scoring quality versus expert baseline, and demonstrated carbon tracking accuracy against actual emission measurements.

JUDGING CRITERIA

Criteria	Weight
Innovation	25%
Business Impact	25%
Technical Excellence	20%
Scalability	15%
User Experience	15%

4 AI Intelligence Platform for Data Centre EPC Project Delivery

Theme: *Industrial Intelligence / Infrastructure Construction / Quality Management*

PROBLEM CONTEXT

India is in the middle of a data centre construction boom. JLL's 2025 India Data Centre Report projects national capacity growing from approximately 900 MW in 2024 to over 2,700 MW by 2027, representing capital deployment of $15 billion or more. A single hyperscale facility involves 15,000 to 40,000 equipment line items, up to 200 concurrent trade contractors, and commissioning sequences spanning thousands of individual test procedures across power, cooling, and IT infrastructure — all with zero tolerance for errors that would compromise the future uptime SLA. Yet a 2024 Turner & Townsend survey found that 67% of data centre EPC projects in Asia-Pacific experienced schedule overruns exceeding 10%, with procurement misalignment and commissioning failures as the leading causes. The underlying problem is information fragmentation: specifications, vendor submittals, test records, RFI logs, and change orders exist in disconnected systems, and the intelligence to connect them never gets built. For India to realise its ambitions as an AI infrastructure hub, it needs EPC delivery capability that matches the complexity of what it is building. That requires an AI layer over project data — one that can catch specification deviations before they reach site, predict schedule risks weeks ahead, and ensure the as-built facility meets the quality standards required for Tier III and IV certification.

CHALLENGE STATEMENT

Build an AI-powered EPC Project Intelligence platform for data centre construction that unifies project documents, specifications, schedules, procurement data, and quality records into a living intelligence layer — enabling proactive schedule management, automated compliance and quality checking, and real-time commissioning support across the full project lifecycle.

WHAT YOU MAY BUILD

Participants may explore areas such as:

- **Specification & Quality Compliance Agent —** AI agent that ingests thousands of pages of equipment specifications, design standards, and client requirements — and automatically checks procurement orders, vendor submittals, and shop drawings for deviations, flagging non-conformances before they reach site and logging them against the quality management audit trail.
- **Predictive Schedule Risk Engine —** Multi-agent system that analyses project schedules against real-time procurement status, equipment lead times, workforce availability, and weather — identifying critical path risks weeks in advance and generating mitigation options, not just alerts.
- **Supply Chain Visibility & Risk Agent —** Geospatial AI that tracks critical equipment shipments (UPS systems, generators, cooling towers, switchgear) across multi-tier suppliers — alerting teams to at-risk deliveries and modelling procurement alternatives before they become critical path issues.
- **Commissioning Quality Assurance Copilot —** AI agent trained on data centre commissioning standards (TIA-942, BICSI, Uptime Institute Tier specifications) that guides engineers through integrated system testing sequences, auto-generates test records, flags non-conformances against acceptance criteria, and builds the as-commissioned quality documentation package.
- **Project Knowledge & RFI Intelligence Agent —** RAG-powered conversational layer over all project documents — specifications, submittals, RFIs, meeting minutes, change orders —

that answers technical and contractual queries in seconds, with citations, and identifies when similar RFIs have been resolved previously to reduce re-work cycles.

These examples are illustrative only.

SUGGESTED TECHNOLOGIES

- Agentic AI / Multi-Agent Systems
- RAG over large technical document and specification corpora
- Computer Vision (drawing review, submittal checking)
- Geospatial Intelligence (supply chain tracking)
- Quality Management System (QMS) Integration
- Predictive Analytics (schedule and cost risk modelling)

EXPECTED DELIVERABLES

- Working Prototype
- Architecture Diagram
- Presentation Deck
- Demo Video

Evaluation Focus Specification compliance detection accuracy on test cases, schedule risk prediction lead time versus actual delays, supply chain visibility depth and alerting timeliness, commissioning test automation coverage, and demonstrated reduction in manual coordination effort — measured in hours, not percentages.

JUDGING CRITERIA

Criteria	Weight
Innovation	25%
Business Impact	25%
Technical Excellence	20%
Scalability	15%
User Experience	15%

5 AI-Powered Urban Air Quality Intelligence for Smart City Intervention

Theme: *Smart Cities / Environmental Intelligence / Geospatial Analytics / Public Health*

PROBLEM CONTEXT

India's air quality crisis is not a Delhi problem — it is a national urban crisis. In 2024-25, Delhi averaged an AQI of 218 (classified 'Poor' or worse for over 200 days), but the situation across other metros is nearly as severe: Mumbai recorded dangerous AQI levels on over 60 days in 2024, Kolkata averaged AQI above 150 for large parts of the winter season, and Bengaluru and Chennai — long considered relatively clean cities — have seen measurable deterioration as vehicle density and construction activity surge. CPCB's National Air Quality data for 2024 shows that 24 of India's 50 most polluted cities are Tier 1 or Tier 2 urban centres. The Lancet Planetary Health journal estimated 1.67 million premature deaths annually from air pollution in India — a public health burden that falls disproportionately on urban populations. Despite India deploying over 900 Continuous Ambient Air Quality Monitoring Stations (CAAQMS) under the National Clean Air Programme, a 2024 CAG audit found that only 31% of cities with monitoring data had any actionable multi-agency response protocols linked to those readings. The data exists. The intelligence layer to act on it does not. City administrations need more than dashboards. They need geospatial attribution (which sources are responsible at this location, right now), predictive forecasting (what will AQI be in 24 hours at ward level), and enforcement intelligence (where to deploy inspectors for maximum impact). That combination does not exist today.

CHALLENGE STATEMENT

Build an AI-powered Urban Air Quality Intelligence platform that fuses monitoring station data, satellite imagery, mobility feeds, meteorological forecasts, and geospatial land use layers to move from reactive monitoring to proactive, evidence-based intervention — giving city administrators the tools to reduce pollution at source rather than just measure it.

WHAT YOU MAY BUILD

Participants may explore areas such as:

- **Geospatial Pollution Source Attribution Engine —** Multi-modal AI agent that analyses spatial-temporal AQI patterns against land use maps, traffic density, construction permits, industrial stacks, and satellite-detected thermal anomalies — attributing pollution by source category at ward or zone level with statistical confidence scores.
- **Hyperlocal Predictive AQI Forecasting Agent —** AI model integrating meteorological forecasts, traffic prediction, seasonal emission calendars, and atmospheric dispersion modelling to provide 24-72 hour AQI forecasts at 1km grid resolution across city boundaries — enabling intervention scheduling rather than reactive advisories.
- **Enforcement Intelligence & Prioritisation Agent —** Agent that correlates pollution hotspot data with registered emission sources — industries, construction sites, waste burning locations, diesel fleet movement — and generates prioritised, evidence-backed enforcement action recommendations for municipal and pollution control authorities with supporting geospatial documentation.
- **Multi-City Comparative Intelligence Dashboard —** Geospatial analytics layer that tracks and compares air quality trends, intervention effectiveness, and compliance metrics across multiple urban centres — enabling city administrators to learn from interventions that worked in comparable cities.

- **Citizen Health Risk Advisory System —** AI platform that generates ward-level health risk alerts, maps population vulnerability (hospitals, schools, outdoor workers, elderly populations) against forecast AQI, and pushes personalised advisories through mobile apps, public displays, and IVR in regional languages — Bengaluru in Kannada, Chennai in Tamil, and so on.

These examples are illustrative only.

SUGGESTED TECHNOLOGIES

- Geospatial Intelligence & Remote Sensing (Sentinel satellite, MODIS)
- Multi-Agent AI Systems
- Real-Time IoT Sensor Data Integration (CAAQMS)
- Atmospheric Dispersion Modelling
- Predictive Analytics
- LLMs for multi-language citizen communication

EXPECTED DELIVERABLES

- Working Prototype
- Architecture Diagram
- Presentation Deck
- Demo Video

Evaluation Focus Source attribution accuracy versus ground-truth emission inventories, AQI forecast accuracy at hyperlocal resolution (RMSE versus persistence baseline), enforcement recommendation quality rated by domain experts, citizen advisory relevance and language coverage, and demonstrated reduction in response time from signal to intervention.

JUDGING CRITERIA

Criteria	Weight
Innovation	25%
Business Impact	25%
Technical Excellence	20%
Scalability	15%
User Experience	15%

6 AI for Digital Public Safety: Defeating Counterfeiting, Fraud & Digital Arrest Scams

Theme: *Smart Cities / Public Safety / Digital Trust / Geospatial Law Enforcement*

PROBLEM CONTEXT

India registered 1.14 million cybercrime complaints in 2023 — up 60% from 2022 — and the trajectory has only steepened. The Ministry of Home Affairs reported that 'digital arrest' scams, where fraudsters impersonating CBI, ED, or Customs officers trap victims in multi-day psychological hostage situations over video call, defrauded citizens of over Rs 1,776 crore in just the first nine months of 2024. These are not opportunistic crimes — they are industrialised operations run from fraud compounds, often across borders, using spoofed numbers, AI-generated voices, and fake government portals. Separately, counterfeit currency remains a persistent threat: the RBI's Annual Report 2025 flagged record FICN (Fake Indian Currency Notes) seizures, with high-denomination Rs 500 fakes of sufficient quality to defeat manual detection in routine banking operations. What law enforcement lacks is not evidence after the fact — it is intelligence before mass victimisation occurs, and reliable tools to detect threats at the point of contact rather than the point of complaint. Solving this requires convergence of financial transaction intelligence, communication network analysis, physical security (counterfeit detection), and real-time public safety coordination — exactly the kind of multi-source, multi-agency intelligence problem where AI has the potential to be transformative.

CHALLENGE STATEMENT

Build an AI-powered Digital Public Safety Intelligence platform that equips law enforcement agencies, financial institutions, and citizens with proactive tools to detect, disrupt, and respond to digital fraud networks, counterfeit currency circulation, and organised scam operations — shifting from reactive case investigation to predictive threat neutralisation.

WHAT YOU MAY BUILD

Participants may explore areas such as:

- **Digital Arrest Scam Detection & Alerting —** Real-time AI classifier trained on digital arrest scam patterns — call flow sequences, number spoofing signatures, script templates, video call metadata — that flags active scam sessions to telecom providers and potential victims before financial transfer occurs, with automated MHA alert generation.
- **Counterfeit Currency Identification Agent —** Computer vision AI deployable on mobile devices, bank counting machines, and point-of-sale terminals that identifies fake notes through microprint analysis, security thread verification, serial number pattern validation, and UV feature simulation — providing field officers and bank tellers with instant, reliable identification across all denominations.
- **Fraud Network Graph Intelligence —** Graph AI agent that analyses transaction metadata, call records, device fingerprints, and account linkages to map coordinated fraud campaigns — clustering victim reports, scammer infrastructure, and money mule networks into actionable intelligence packages that link across jurisdictions and can be submitted as court-admissible evidence.
- **Geospatial Crime Pattern Intelligence —** Geospatial AI layer for law enforcement that maps fraud complaint locations, counterfeit currency seizure points, and cybercrime hotspots — enabling patrol prioritisation, resource deployment, and inter-district intelligence sharing in near real time through a command centre interface.
- **Citizen Fraud Shield (Multi-channel) —** Conversational AI accessible via WhatsApp, IVR, and mobile app that walks citizens through real-time fraud risk assessment for suspicious calls,

payment requests, or messages — providing instant verdicts, guided reporting to NCRB portals, and advisory in 12 regional languages.

These examples are illustrative only.

SUGGESTED TECHNOLOGIES

- Computer Vision (counterfeit detection, deepfake identification)
- Graph AI & Network Analysis (fraud ring mapping)
- NLP / LLMs (scam script and voice pattern classification)
- Geospatial Intelligence (crime mapping, patrol optimisation)
- Speech AI (voice spoofing and AI-voice detection)
- Agentic AI for multi-source intelligence fusion

EXPECTED DELIVERABLES

- Working Prototype
- Architecture Diagram
- Presentation Deck
- Demo Video

Evaluation Focus Counterfeit detection accuracy across denominations and print quality, digital arrest scam detection precision and recall, fraud network detection lead time before mass victimisation, false positive rate for citizen-facing tools (must be very low), and auditability of intelligence packages for legal admissibility.

JUDGING CRITERIA

Criteria	Weight
Innovation	25%
Business Impact	25%
Technical Excellence	20%
Scalability	15%
User Experience	15%

7 AI-Driven Cyber Resilience for Critical National Infrastructure

Theme: *Cybersecurity / Industrial Intelligence / National Security*

PROBLEM CONTEXT

Critical national infrastructure has become a primary target for sophisticated cyber actors — and India's public institutions have borne the brunt of this over the past two years. CERT-In reported handling over 1.59 million cybersecurity incidents in 2023, a figure that has continued to climb through 2024-25. The pattern of high-profile attacks is deeply concerning: AIIMS Delhi was paralysed for over two weeks by ransomware in 2022; in 2024, CBSE suffered a data breach affecting examination records; and in early 2026, a coordinated cyberattack targeted CBSE's digital infrastructure ahead of board examinations, compromising student data and forcing emergency system shutdowns across multiple states — an incident that exposed how exposed government education systems remain despite repeated warnings. India's National Cyber Security Policy acknowledges that over 70% of government entities operate on end-of-life IT infrastructure, meaning attackers are not working hard to find entry points. The deeper problem is detection speed. Most public sector organisations discover breaches only after significant damage has occurred — weeks or months after initial infiltration. Advanced persistent threats (APTs) deliberately operate at low-and-slow speeds designed to evade signature-based detection. What is needed is a behavioural intelligence layer that detects anomalies from how systems normally behave, not from whether they match a known malware signature — because by the time a signature exists, the attack has already succeeded somewhere.

CHALLENGE STATEMENT

Build an AI-powered Cyber Resilience platform for critical national infrastructure that autonomously detects behavioural anomalies, correlates weak signals across heterogeneous IT and OT environments, maps attack progression against known threat frameworks, and orchestrates containment actions — compressing the time from initial compromise to detection and response from weeks to hours.

WHAT YOU MAY BUILD

Participants may explore areas such as:

- **Behavioural Anomaly Detection Engine —** Multi-agent AI system that builds baseline behavioural profiles for users, devices, and network segments — then continuously scores deviations from those baselines across log data, network flows, and endpoint telemetry, without relying on known malware signatures.
- **APT Campaign Attribution & Prediction Agent —** AI agent trained on MITRE ATT&CK framework, CERT-In advisories, and threat actor TTPs that maps observed attack patterns to known campaigns, predicts likely next-stage moves, and generates specific defensive actions tailored to the organisation's architecture.
- **Autonomous Incident Response Orchestrator —** AI-driven SOAR layer that executes pre-approved containment playbooks — isolate endpoint, revoke credential, block IP, snapshot VM state — within seconds of high-confidence threat confirmation, with human escalation gates for decisions above defined blast radius thresholds.
- **Government Infrastructure Vulnerability Prioritisation —** AI agent that continuously maps the organisation's asset inventory against live CVE feeds, contextualises exploitability given the specific network topology and observed threat actor profiles, and generates a dynamic, risk-ranked remediation queue — addressing the reality that government teams cannot patch everything at once.

- **Cyber Resilience Digital Twin —** AI-generated simulation of the organisation's security architecture that enables attack path modelling, red team scenario testing, and impact assessment of proposed security investments — all without touching live production systems.

These examples are illustrative only.

SUGGESTED TECHNOLOGIES

- Agentic AI / Multi-Agent Systems
- Unsupervised Anomaly Detection (user and entity behaviour analytics)
- Graph AI (attack path analysis, lateral movement detection)
- RAG over threat intelligence, CVE databases, and CERT-In advisories
- Knowledge Graphs (MITRE ATT&CK TTP mapping)
- SOAR Integration & Response Automation

EXPECTED DELIVERABLES

- Working Prototype
- Architecture Diagram
- Presentation Deck
- Demo Video

Evaluation Focus Anomaly detection rate and false positive rate on benchmark datasets, APT attribution accuracy at MITRE ATT&CK technique level, incident response automation coverage (percentage of playbook steps executable autonomously), MTTD/MTTR improvement versus baseline SOC, and full auditability of every automated action taken.

JUDGING CRITERIA

Criteria	Weight
Innovation	25%
Business Impact	25%
Technical Excellence	20%
Scalability	15%
User Experience	15%

8 AI for Industrial Knowledge Intelligence: Unified Asset & Operations Brain

Theme: *Industrial Intelligence / Document Management / Knowledge Engineering / Quality*

PROBLEM CONTEXT

A 2024 McKinsey global survey found that professionals in asset-intensive industries spend an average of 35% of their working hours searching for information, clarifying instructions, or recreating documents that already exist somewhere in the organisation. In India specifically, a NASSCOM-EY study of manufacturing and energy companies found that the average large plant operates across 7 to 12 disconnected document systems — P&IDs and engineering drawings in one place, maintenance work orders in another, operating procedures in a third, inspection records in a fourth, and regulatory submissions scattered across email archives. BIS Research estimated that this fragmentation contributes to 18–22% of unplanned downtime events in Indian heavy industry, as maintenance teams make decisions without complete equipment history or failure pattern context. Then there is the knowledge cliff: an estimated 25% of India's experienced industrial engineers and operators will retire within the next decade, taking decades of undocumented operational knowledge with them. Once gone, it cannot be recovered. Knowledge fragmentation in industrial operations is not a file management problem. It is a safety problem, a quality problem, and an operational efficiency problem — and it compounds over time. The organisations that solve it first will have a structural advantage in how they operate, maintain, and improve their assets.

CHALLENGE STATEMENT

Build an AI-powered Industrial Knowledge Intelligence platform that ingests heterogeneous documents — engineering drawings, maintenance records, safety procedures, inspection reports, operating instructions, project files — across structured and unstructured formats, and makes their collective intelligence queryable, actionable, and continuously updated at the point of need, across any device or function.

WHAT YOU MAY BUILD

Participants may explore areas such as:

- **Universal Document Ingestion & Knowledge Graph Agent —** AI pipeline that processes PDFs, P&IDs, scanned forms, spreadsheets, and email archives — extracting entities (equipment tags, process parameters, regulatory references, personnel, dates) and building a unified knowledge graph that maintains relationships across document types and updates automatically as new records arrive.
- **Expert Knowledge Copilot —** RAG-powered conversational AI that answers operational, maintenance, and engineering queries across the full document corpus — with source citations, confidence scores, and direct links to the originating documents. Built to work on mobile for field technicians, not just desktops for engineers.
- **Maintenance Intelligence & RCA Agent —** AI agent that fuses work order history, equipment failure records, OEM manuals, inspection findings, and real-time operating conditions to generate predictive maintenance recommendations, Root Cause Analysis (RCA) support, and optimised maintenance schedules — reducing unplanned downtime by connecting the dots that no individual team member can connect alone.
- **Quality & Regulatory Compliance Intelligence —** Agentic system that maps regulatory requirements (Factory Act, OISD, PESO, environmental norms, quality standards) against current procedures, equipment states, and inspection records — identifying compliance

gaps, auto-generating compliance evidence packages for audits, and flagging quality deviations before they escalate.

- **Lessons Learned & Failure Intelligence Engine —** AI agent that analyses incident reports, near-miss records, audit findings, and quality non-conformances across the organisation's history and external industry databases — identifying systemic patterns invisible to any individual review, and proactively pushing relevant warnings to operational teams before similar conditions recur.

These examples are illustrative only.

SUGGESTED TECHNOLOGIES

- RAG (Retrieval-Augmented Generation) over heterogeneous industrial document corpora
- Knowledge Graphs & Industrial Ontology Engineering
- Computer Vision (P&ID parsing, drawing digitisation)
- OCR & Document Intelligence (structured + unstructured)
- Quality Management System (QMS) Integration
- Agentic AI for maintenance and compliance workflows

EXPECTED DELIVERABLES

- Working Prototype
- Architecture Diagram
- Presentation Deck
- Demo Video

Evaluation Focus Entity extraction accuracy across document types, query answer quality on domain-expert benchmark questions, knowledge graph linkage completeness, time-to-answer versus traditional search, compliance gap detection accuracy, and demonstrated improvement in cross-functional knowledge discovery — ideally validated with real industrial document samples.

JUDGING CRITERIA

Criteria	Weight
Innovation	25%
Business Impact	25%
Technical Excellence	20%
Scalability	15%
User Experience	15%